Library Management System — User Guide

From Harel Rajendram
Id: 2534297

This is an explanation of all the features provided from this library system

Section 1 — Program Startup

The home screen when opening the system

```
/Users/macos/Library/Java/JavaVirtualMachines/openjdk-25/Contents/Home/bin/java -javaagent:/Users/macos/Desktop/IntelliJ IDEA

===== LIBRARY SYSTEM MENU =====
1. Search
2. Borrow Item
3. Return Item
4. Mark Item as LOST
5. Admin Menu
6. Exit
Choose an option: 1
```

Section 2 — Search by Title

You can find the book based off the title

```
===== SEARCH MENU =====
1. Search by Title
2. Search by Title (Recursive)
3. Search by Author
4. Search by Author (Recursive)
5. Back
Choose an option: 1
Enter keyword: 1984
[]
```

Section 3 — Recursive Search

Different methods to search for your books like recursion

```
===== LIBRARY SYSTEM MENU =====
1. Search
2. Borrow Item
3. Return Item
4. Mark Item as LOST
5. Admin Menu
6. Exit
Choose an option: 1

===== SEARCH MENU =====
1. Search by Title
2. Search by Title (Recursive)
3. Search by Author
4. Search by Author (Recursive)
5. Back
Choose an option: 2
Enter keyword: Farenheit
[]
```

Section 4 — Borrow Item

You can borrows book from the library system by searching its name and Id

```
===== LIBRARY SYSTEM MENU =====
1. Search
2. Borrow Item
3. Return Item
4. Mark Item as LOST
5. Admin Menu
6. Exit
Choose an option: 2
Enter User ID: S1
Enter Item ID: 42
Error: Item does not exist

===== LIBRARY SYSTEM MENU =====
1. Search
2. Borrow Item
3. Return Item
4. Mark Item as LOST
5. Admin Menu
6. Exit
Choose an option: 2
Enter User ID: S1
Enter Item ID: 103
Item borrowed successfully.
```

Section 5 — Borrow Error

If the book you are trying to borrow is already borrowed by someone else, it will send a message saying it is borrowed

```
===== LIBRARY SYSTEM MENU =====
1. Search
2. Borrow Item
3. Return Item
4. Mark Item as LOST
5. Admin Menu
6. Exit
Choose an option: 2
Enter User ID: S1
Enter Item ID: 103
Error: Item already borrowed
```

Section 6 — Return Item

You are able to return the books that you have borrowed

```
===== LIBRARY SYSTEM MENU =====
1. Search
2. Borrow Item
3. Return Item
4. Mark Item as LOST
5. Admin Menu
6. Exit
Choose an option: 3
Enter User ID: S1
Enter Item ID: 103
Item returned successfully.
```

Section 7 — Mark Lost

The library system can mark items as lost

```
===== LIBRARY SYSTEM MENU =====
1. Search
2. Borrow Item
3. Return Item
4. Mark Item as LOST
5. Admin Menu
6. Exit
Choose an option: 4
Enter Item ID to mark LOST: 102
Item marked as LOST.
```

Section 8 — Admin Report

Admin users can access the admin report will lets them see more information

```
===== LIBRARY SYSTEM MENU =====
1. Search
2. Borrow Item
3. Return Item
4. Mark Item as LOST
5. Admin Menu
6. Exit
Choose an option: 5

===== ADMIN MENU =====
1. Generate Report
2. Add New Book
3. Backup Data
4. Back
Choose an option: 1

===== ADMIN REPORT =====

Borrowed Items:

In-Store Items:
org.yaa.Book@6ce253f1
org.yaa.Book@53d8d10a
org.yaa.Book@e9e54c2
org.yaa.Book@65ab7765

Lost Items:
org.yaa.DVD@7229724f
```

Section 9 — Add New Book

You are able to add new books that the library does not have to its lists of books

```
===== LIBRARY SYSTEM MENU =====
1. Search
2. Borrow Item
3. Return Item
4. Mark Item as LOST
5. Admin Menu
6. Exit
Choose an option: 5

===== ADMIN MENU =====
1. Generate Report
2. Add New Book
3. Backup Data
4. Back
Choose an option: 2
Enter Book ID: 200
Enter ISBN: 9781234567890
Enter Title: Clean
Enter Author: Yi Wang
Enter Genre (SCIENCE, SCI_FI, HISTORY, FANTASY): SCIENCE
Book added successfully. New ID: 200
```

Section 10 — Backup

This is the backup data of the library

```
===== LIBRARY SYSTEM MENU =====
1. Search
2. Borrow Item
3. Return Item
4. Mark Item as LOST
5. Admin Menu
6. Exit
Choose an option: 5

===== ADMIN MENU =====
1. Generate Report
2. Add New Book
3. Backup Data
4. Back
Choose an option: 3
Backup completed successfully.
```